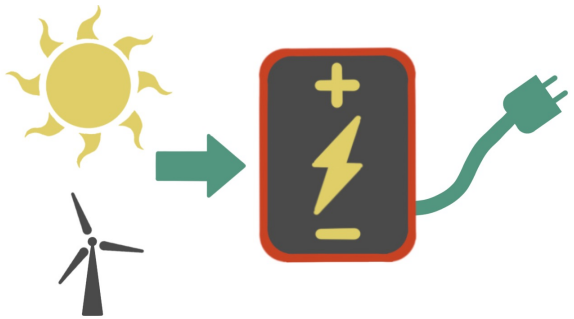
Reliability and accelerated lifetime testing



Ania's BMS Course

by Ania Mitros, PhD

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https://ofb.net/~ania/teaching/BMS-course/

Reliability

Video 1
- Introduction: Lifetime and accelerated lifetime testing
- Arrhenius equation
  - Review Arrhenius equation
  - Derive acceleration factor
  - AEC-Q100

Video 2
- Caveats of the approach
- Mission profile
- The design and test process

Over-design: Reduced profit margin on every unit.
Under-design: Warranty costs.

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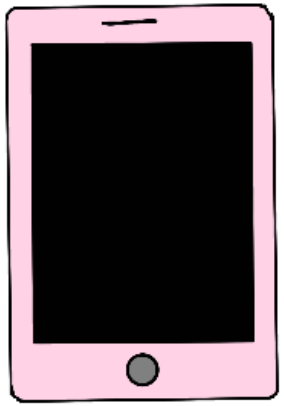
Application lifetime

Design lifetime: Target for the minimum time that all units work.

(Not a target for failure.)

Skateboard

2-3 yrs

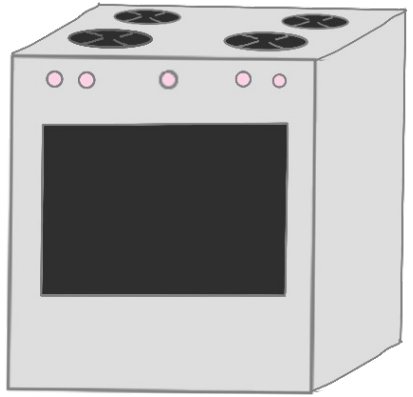


Phone

iPhone: 4-8 yrs
Samsung: 3-6 yrs
Oppo: 2-3 yrs

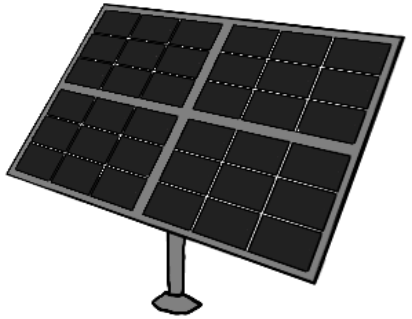
Car

8 years



Home appliance

Microwave: 9 yrs
Dishwasher: 9 yrs
Refrigerator: 13 yrs
Dryer: 13 yrs
Gas range: 15 yrs
Tesla Powerwall: 10 yrs



Solar panels

25-30 yrs
(estimates vary)

Inverters: 10-15 yrs

Phone lifetime: https://everphone.com/en/blog/smartphone-lifespan/

Appliance lifetime: https://www.thisoldhouse.com/21018361/how-long-things-last

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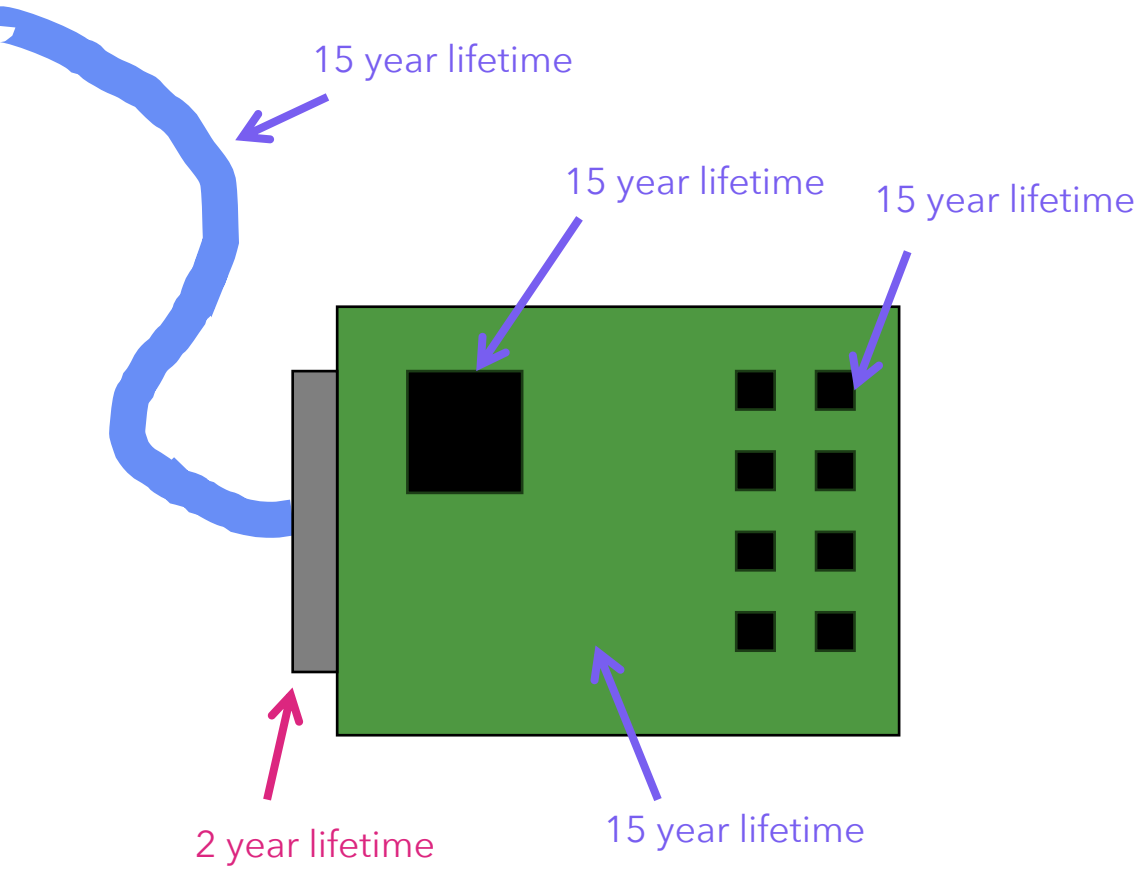
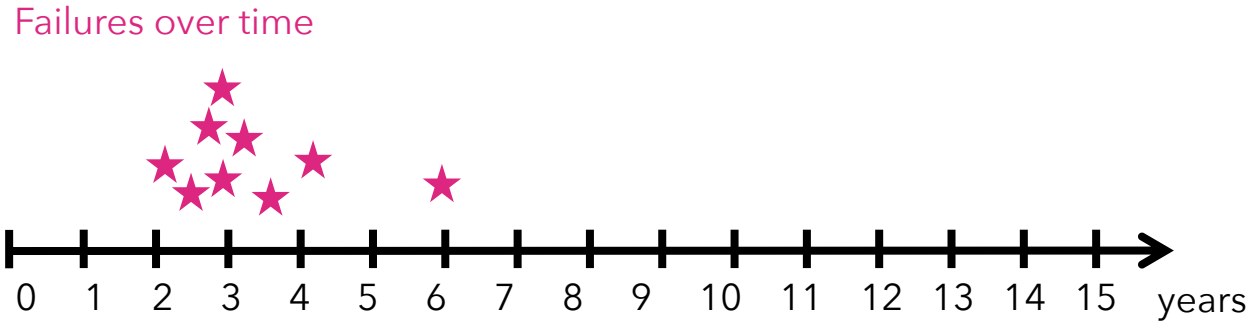
Lifetime is minimum of component lifetimes



Lifetime of least reliable component is product lifetime

Unless:

- The component is replaceable
- The component is redundant
- Other designed mitigation



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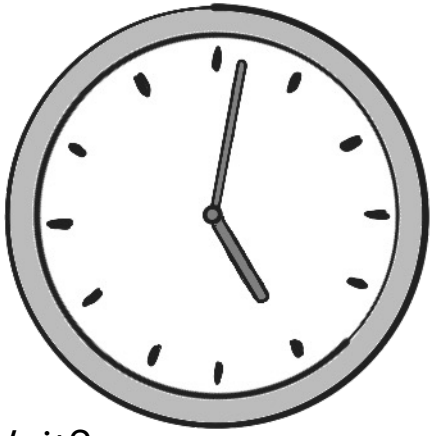
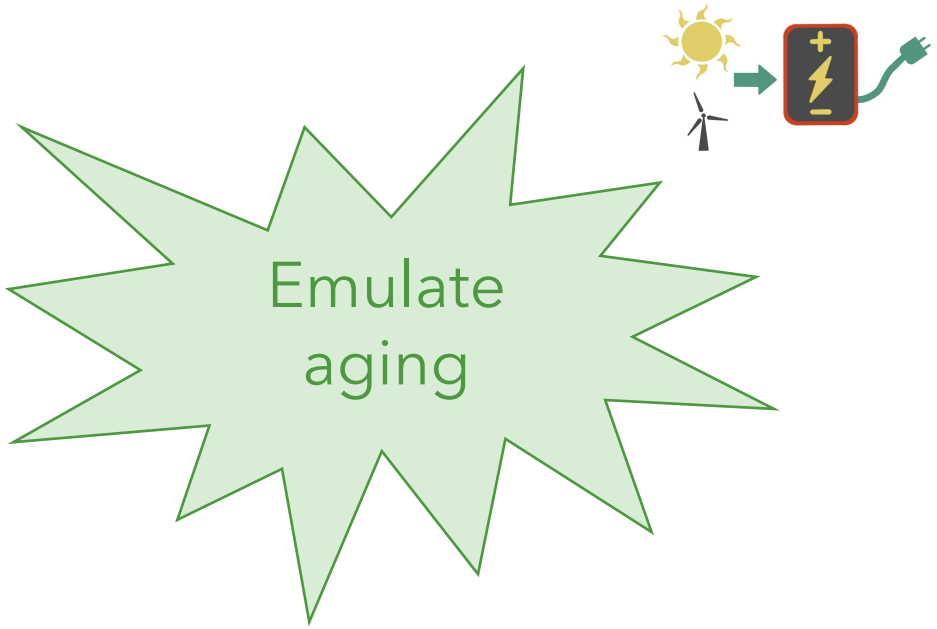
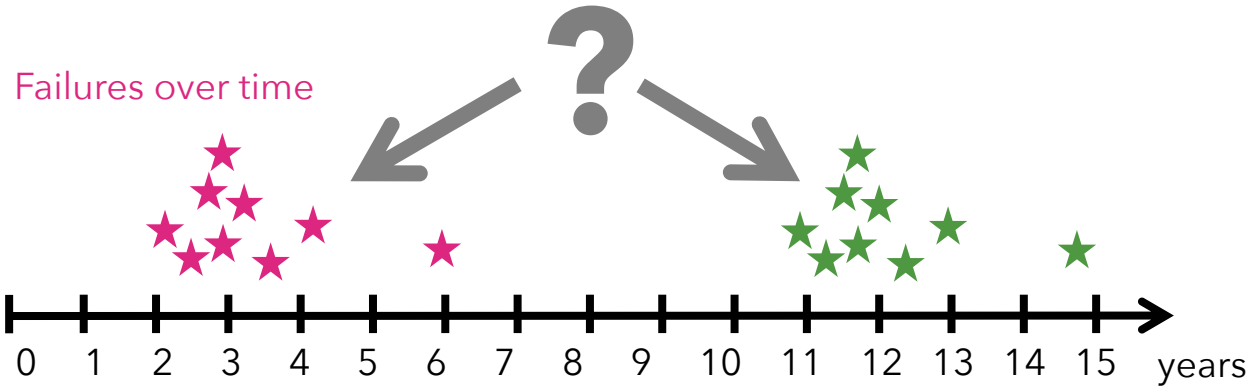
How to test for reliability?



Wait?
→ 4-8 years to release next iPhone
→ 8 years to release next car model
→ 15 years to release next gas range
Too long!



How to verify whether the failures over time fit our target?



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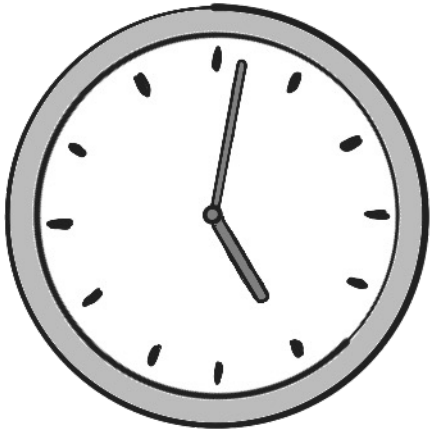
How to test for reliability?



Emulate aging by increasing temperature

…in ice water | Dissolve sugar in water | …in hot tea

Sugar dissolves faster in warmer water

because chemical reactions occur faster at higher temperatures

If aging is due to chemical reactions, temperature accelerates aging

Arrhenius equation

$$k = A \cdot e^{\frac{-E_a}{k_B T}}$$

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Summary

- Application lifetimes vary, with corresponding reliability needs
- Increasing temperature
  - Increases reaction rate
    - Accelerates many aging mechanisms
- If we model this well, we can emulate accelerated aging

  premise behind accelerated lifetime testing

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Arrhenius Equation

$$k = A \cdot e^{\frac{-E_a}{k_B T}}$$

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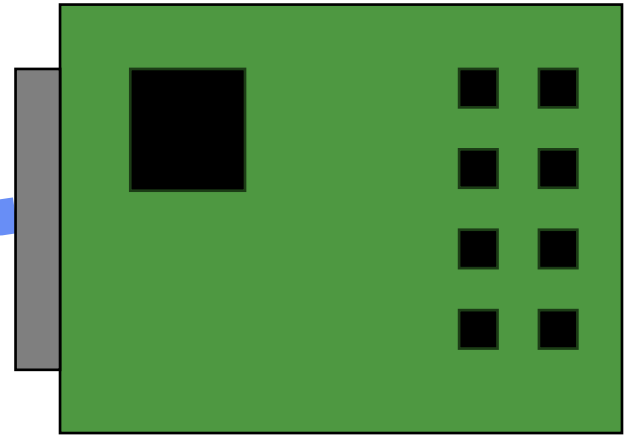
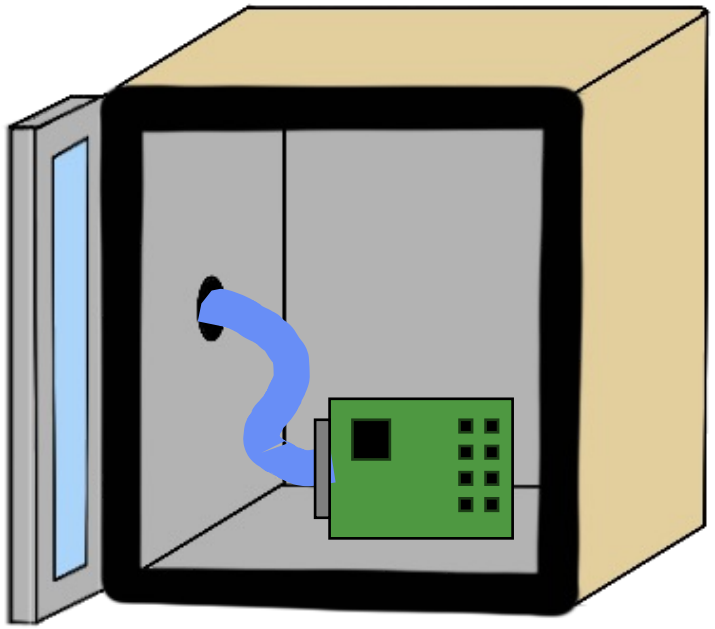
Example of using Arrhenius Equation

Product: **15 year lifetime**,
87°C average temperature,
2 hrs/day → 12,000 hrs lifetime



Thermal chamber for
accelerated lifetime testing



Actually, we'll
bake many units.

How long to bake?
→ Arrhenius equation

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Arrhenius Equation

Quantifies rate by which reactions are accelerated by temperature

Conceptually: The number of effective collisions is proportional to the number of molecules above a certain threshold energy

Rate constant
Number of molecules colliding per second and resulting in a reaction

$$k = A \cdot e^{\frac{-E_a}{k_B T}}$$

Probability of a given collision resulting in a reaction

Number of collisions per second

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Arrhenius Equation

Quantifies rate by which processes are accelerated by temperature

Rate of collision between molecules resulting in a reaction

How fast the process proceeds

k : rate constant

Varies by failure mechanism

E_a : Activation Energy

$$k = A \cdot e^{\frac{-E_a}{k_B T}}$$

T : Temperature in Kelvin

k_B : Boltzmann constant = 8.62x10^{-5} eV/K

e : exponential function

A : pre-exponential factor

Experimentally fit

...and not needed for accelerated lifetime testing

Engineers use:
k_B = 8.62x10^{-5} eV/K
Physicists use:
k_B = 1.38x10^{-23} J/K

Chemists replace k_B with R = 8.314 J/mol K and correspondingly adjust units for E_a

$$k = A \cdot e^{\frac{-E_a}{RT}}$$

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Arrhenius Equation

Don't confuse k and k_B

k : rate constant

$$k = A \cdot e^{\frac{-E_a}{k_B T}}$$

k_B : Boltzmann constant = 8.62×10^{-5} eV/K

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Arrhenius Equation: Rate and temperature

Rate relates exponentially to temperature

How fast the process proceeds

k : rate constant

$$k = A \cdot e^{\frac{-E_a}{k_B T}} = A \cdot \frac{1}{e^{\frac{E_a}{k_B T}}}$$

$+\Delta T \rightarrow$ exponential increase in k

T : Temperature in Kelvin

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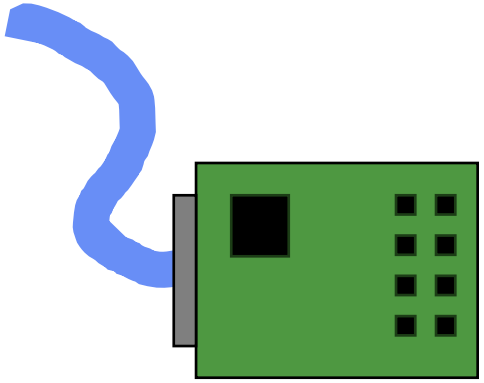
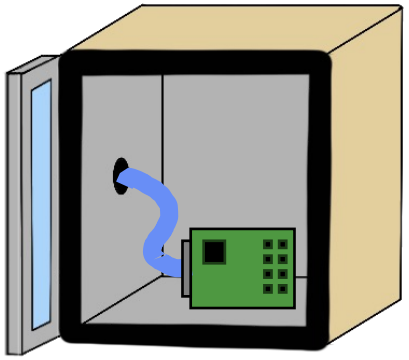
Arrhenius Equation applied

How much acceleration from a given ΔT (temperature)?





Product:
15 year lifetime
Used 2hrs/day
→ (2×365.25×15) = 12,000 hours of operating use time over lifetime
Ave temp: T_{app}=87°C

→ Bake at: T_{bake} > 87°C

What temperature?

Bake how long?

From:

$$k = A \cdot e^{\frac{-E_a}{k_B T}}$$

Find acceleration factor:

$$A_T = \frac{k_{bake}}{k_{app}} = \frac{t_{app}}{t_{bake}}$$

A_T quantifies how much faster aging proceeds during baking than during normal operation.

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Define the acceleration factor

Express with bake time t_{bake} and application lifetime t_{app}

$$A_T = \frac{k_{bake}}{k_{app}} = \frac{t_{app}}{t_{bake}}$$

If you rate **doubles**
then time **halves**

$$\text{rate} = \frac{1}{time}$$

Equation for bake time t_{bake}

$$t_{bake} = t_{app} \times A_T$$

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Acceleration factor from Arrhenius Equation

For the application:

$$k_{app} = A \cdot e^{\frac{-E_a}{k_B T_{app}}}$$

For the bake:

$$k_{bake} = A \cdot e^{\frac{-E_a}{k_B T_{bake}}}$$

$$A_T = \frac{k_{bake}}{k_{app}} = \frac{A \cdot e^{\frac{-E_a}{k_B T_{app}}}}{A \cdot e^{\frac{-E_a}{k_B T_{bake}}}}$$

$$A_T = e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}$$

Acceleration factor

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Acceleration factor from Arrhenius Equation

For the application:

$$k_{app} = \boxed{A \cdot} e^{\frac{-E_a}{k_B T_{app}}}$$

A cancels out and does not appear in equation for acceleration factor A_T

For the bake:

$$k_{bake} = A \cdot e^{\frac{-E_a}{k_B T_{bake}}}$$

$$A_T = \frac{k_{bake}}{k_{app}} = \frac{A \cdot e^{\frac{-E_a}{k_B T_{app}}}}{A \cdot e^{\frac{-E_a}{k_B T_{bake}}}}$$

$$\boxed{A_T = e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}}$$

E_a : Specific to a particular failure mechanism (ex. 0.7eV)
k_B : Boltzman constant
T_{bake} : Bake temperature
T_{app} : Application operating temperature

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Apply acceleration factor to example

How long and how hot do we need to bake?

$$t_{bake} = t_{app} \times A_T$$

$$t_{bake} = t_{app} \times e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}$$

E_a = 0.7 eV

T_{app} = 87°C

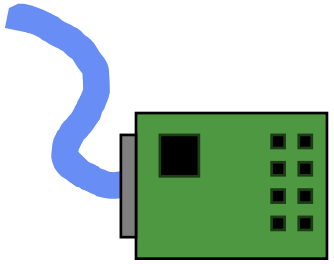
t_{app} = 12000 hrs

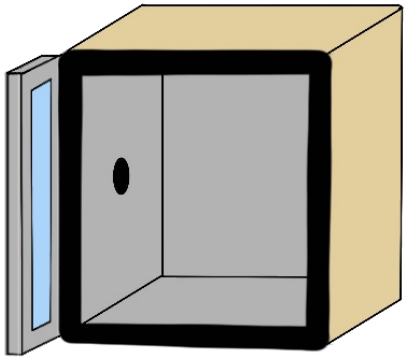
k_B = 8.62x10^{-5} eV/K

T_{bake} °C	T_{bake} K	t_{bake} (hours)
87	360	12000
95	368	7353
105	378	4104
115	388	2361
125	398	1396

3x

9x





15 year lifetime
Used 2hrs/day
→ (2×365.25×15) = 12,000 hours of operating use time over lifetime
Ave temp: T_{app}=87°C

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Source of example: AEC-Q100 – REV-J

AEC-Q100 specifies tests to determine that a device (chip or IC) can be expected to give a level of quality/reliability appropriate for an automotive application.

http://www.aecouncil.com/AECDocuments.html

AEC - Q100 - REV-J
August 11, 2023

Automotive Electronics Council
Component Technical Committee

Table A7.1: Example Basic Calculations for AEC-Q100 Stress Test Conditions and Durations

Loading	Example Mission Profile Input	Stress Test	Stress Conditions	Acceleration Model (all temperatures in K, not in °C)	Model Parameters	Calculated Test Duration	Q100 Test Duration
Operation	t_u = 12,000 hr (average operating use time over 15 yr) T_u = 87°C (average junction temperature in use environment)	High Temperature Operating Life (HTOL)	T_t = 125°C (junction temperature in test environment)	Arrhenius $A_f = \exp\left[\frac{E_a}{k_B} \bullet \left(\frac{1}{T_u} - \frac{1}{T_t}\right)\right]$ Also applicable for High Temperature Storage Life (HTSL) and NVM Endurance, Data Retention Bake, & Operational Life (EDR)	E_a = 0.7 eV (activation energy; 0.7 eV is a typical value, actual values depend on failure mechanism and range from -0.2 to 1.4 eV) k_B = 8.61733 x 10^{-5} eV/K (Boltzmann's Constant)	t_t = 1393 hr (test time) $t_t = \frac{t_u}{A_f}$	1000 hr

Same application assumptions

Same activation energy E_a

Similar result

Common total test time

May pause every few days to check for failures.

AEC-Q100 is for chip (integrated circuits).
AEC-Q200 is for passive components (resistors, capacitors, etc)

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Summary

- Arrhenius equation is exponential in temperature
- Pre-exponential factor A disappears (cancels out) from equation for acceleration factor A_T
- Higher bake temperature T_{bake} allows for shorter test time t_{bake}
- Accelerated lifetime testing based on the Arrhenius equation is widespread
  - The automotive industry standard AEC-Q100 embodies this methodology
  - 1000 hour bake time is common

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Let’s question the approach

Everybody does it. But is it right?

AEC - Q100 - REV-J
August 11, 2023

Automotive Electronics Council
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From: http://www.aecouncil.com/AECDocuments.html

“0.7 eV is a typical value. Actual values depend on failure mechanism and range from -0.2 to 1.4 eV”

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E_a values for silicon semiconductor failures

Component and mechanism	EA (eV)	
MOS devices, including PMOS & CMOS	1.1 – 1.3	transistors
Threshold voltage shift	1.2	transistors
Silicon oxide and Si/SiO_2 interface surface charge accumulation, bipolar	1.0 – 1.05	transistors
Silicon oxide and Si/SiO_2 interface surface charge accumulation, MOS	1.2 – 1.35	transistors
Corrosion (chemical, galvanic, electrolytic)	0.3 – 0.6	
Metallization	1.3– 1.4	Metal (connections)
Intermetallic growth Al/Au	1.0 – 1.05	Metal (connections)
Electromigration	0.5 – 1.2	Metal (connections)
AEC-Q100 “typical”	0.7	

From: Kececioglu, D.B. and Sun, F-B “Burn-in Testing: Its Quantification and Optimization”, 1997
Which cites: Jensen, F. and Peterson, N.E., “Burn-in, An Engineering Approach to the Design and Analysis of Burn-in Procedures”, 1982

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Corrosion (chemical, galvanic, electrolytic)	0.3 – 0.6
Metallization	1.3– 1.4
Intermetallic growth Al/Au	1.0 – 1.05
Electromigration	0.5 – 1.2
AEC-Q100 "typical"	0.7

t_{app} = 12000 hrs

E_a = … eV

$$t_{bake} = t_{app} \times e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}$$

k_B = 8.62x10^{-5} eV/K

T_{app} = 87°C

T_{bake} = 125°C

E_a (eV)	T_{bake} (hours)	T_{bake} (days)
0.3	4773	199
0.5	2581	108
0.7	1396	58
0.9	755	31
1.1	408	17

Imperfect but widespread approach

From: Kececioglu, D.B. and Sun, F-B "Burn-in Testing: Its Quantification and Optimization", 1997
Which cites: Jensen, F. and Peterson, N.E., "Burn-in, An Engineering Approach to the Design and Analysis of Burn-in Procedures", 1982

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Key ideas in “Let’s question the approach”

- **Ea=0.7 eV** is a rough assumption

- **Compromise** between rigor and practicality

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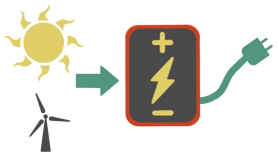
BMS Mission Profile



A mission profile is the collection of relevant environmental and functional loads that a component will be exposed to during its use lifetime. In other words: Customer Specific Lifetime Requirements

From: http://www.aecouncil.com/AECDocuments.html

Temperature, duration, etc

Example Mission Profile

15 year lifetime
Used 2hrs/day
→ (2×365.25×15) = 12,000 hrs of operation over lifetime
Ave temp: T_{app}=87°C

Application lifetimes
2-6 yrs: Small devices
8 yrs: Car
9-15 yrs: Appliances
30 yrs: Solar backup

Car BMS Mission Profile

8 year lifetime
Used 10 hrs/day
→ (10×365.25×8) = 29,220 hrs of operation over lifetime
Ave temp: T_{app}=40 °C

2 hrs driving
8 hrs charging

PCB may be hotter due to passive balancing heat dissipation, or co-packaged power electronics.

Toy BMS Mission Profile

5 year lifetime
Used 4 hrs/day
→ (4×365.25×5) = 7,305 hours of operating use time over lifetime
Ave temp: T_{app}=25 °C

Solar backup BMS Profile

30 year lifetime
Used 16 hrs/day
→ (16×365.25×30) = 175,320 hrs of operation over lifetime
Ave temp: T_{app}=40 °C

4 hrs discharging
12 hrs charging

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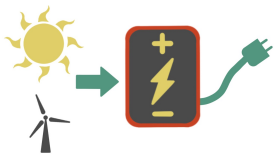
Apply equation to BMS Mission Profiles



From Mission Profiles

	t_{app} (hours)	T_{app} (°C)	t_{bake} (days)
Example	12,000	87°C	58.2
Toy BMS	7,305	25°C	0.3
Car BMS	29,220	40°C	4.8
Solar backup BMS	175,320	40°C	28.8
	175,320	50°C	64.3
	175,320	60°C	136.8
	175,320	70°C	278.2
	175,320	80°C	543.6
	175,320	87°C	849.8

<40 days!

Vary temperature

t_{app} = 12000 hrs

E_a = 0.7 eV

$$t_{bake} = t_{app} \times e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}$$

k_B = 8.62x10^{-5} eV/K

T_{app} = 87°C

T_{bake} = 125°C

Mission profile temperature is key to valid results from accelerated lifetime testing.

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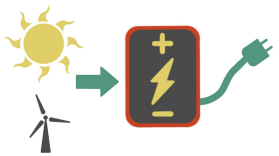
Apply equation… with T_{bake}=85°C



From Mission Profiles

	t_{app} (hours)	T_{app} (°C)	t_{bake} (days)
Example	12,000	87°C	567.1
Toy BMS	7,305	25°C	3.2
Car BMS	29,220	40°C	46.9
Solar backup BMS	175,320	40°C	281.2

Impractical

t_{app} = 12000 hrs

E_a = 0.7 eV

$$t_{bake} = t_{app} \times e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}$$

k_B = 8.62x10^{-5} eV/K

T_{app} = 87°C

T_{bake} = 85°C

Chips designed for 85°C operation may in practice have worse reliability than chips designed for 125°C operation. Many techniques for improving reliability and for increasing max operating temperature are same.

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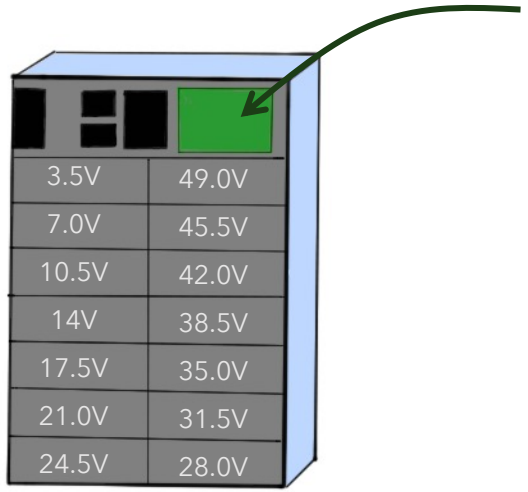
Define a Mission Profile



Max: 40°C

PCB (Printed Circuit Board) with electronics

BMS:

- ASICs:
  - Cell voltage sensing
  - Battery current sensing
  - Cell temperature sensing
- Microcontroller for state estimation
- Support circuits

Power electronics: Switching converter moves energy between the battery's 50V DC and the home's 120V AC.

Heat sources on PCB

Balancing:

- Non-negligible heat
  - 13 * 3.6V * 150mA = 7W
- Short duration

8 kW and 97.5% efficiency:

- 200 W dissipated

(Similar to Tesla Powerwall 3 specs)

Ambient

35°C

daytime

in sunny places where solar energy is most reliable

Up to ~40°C above ambient

Assume 75°C, allowing 35-40°C margin for electronics self-heating.

75°C

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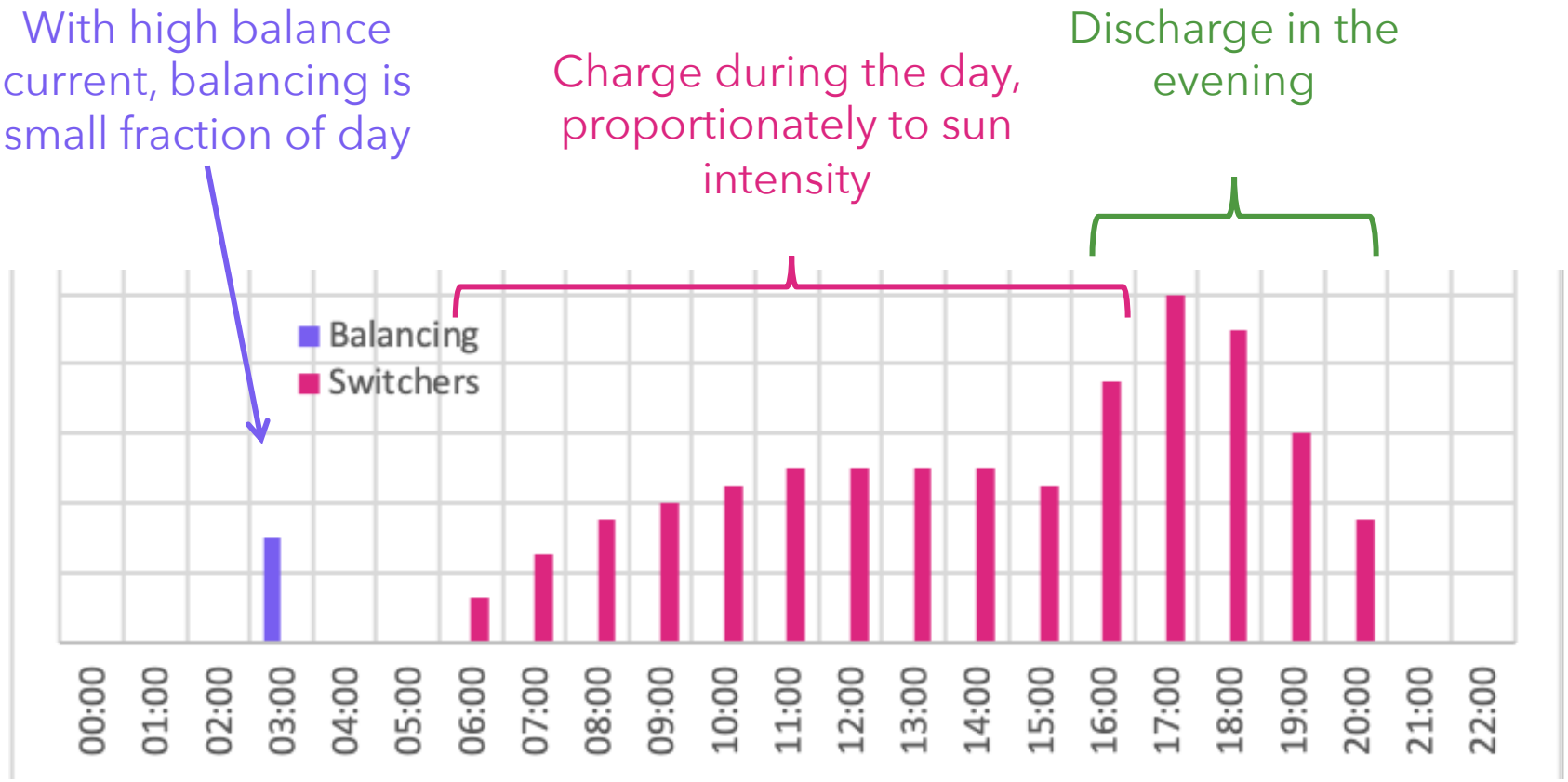
Define a Mission Profile

Daily operating cycle



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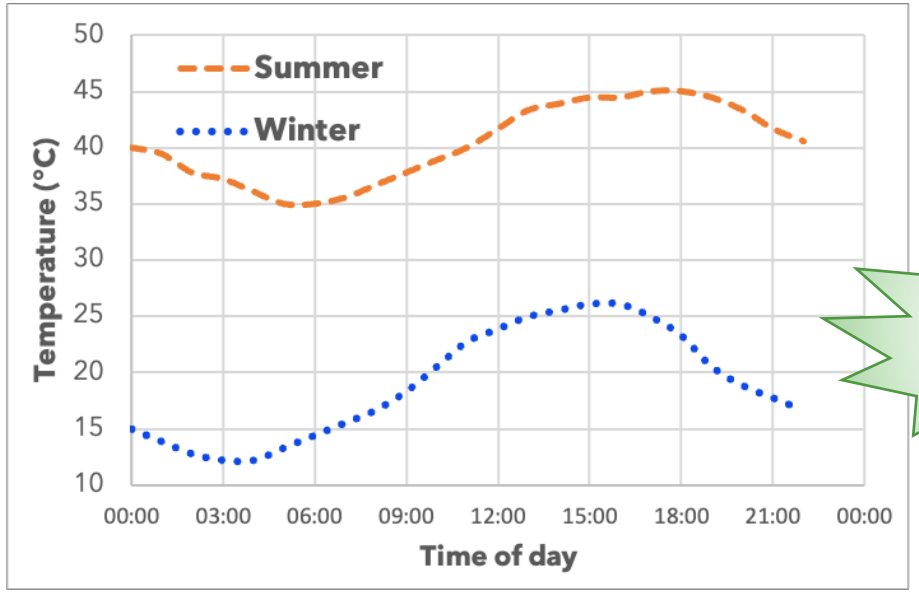
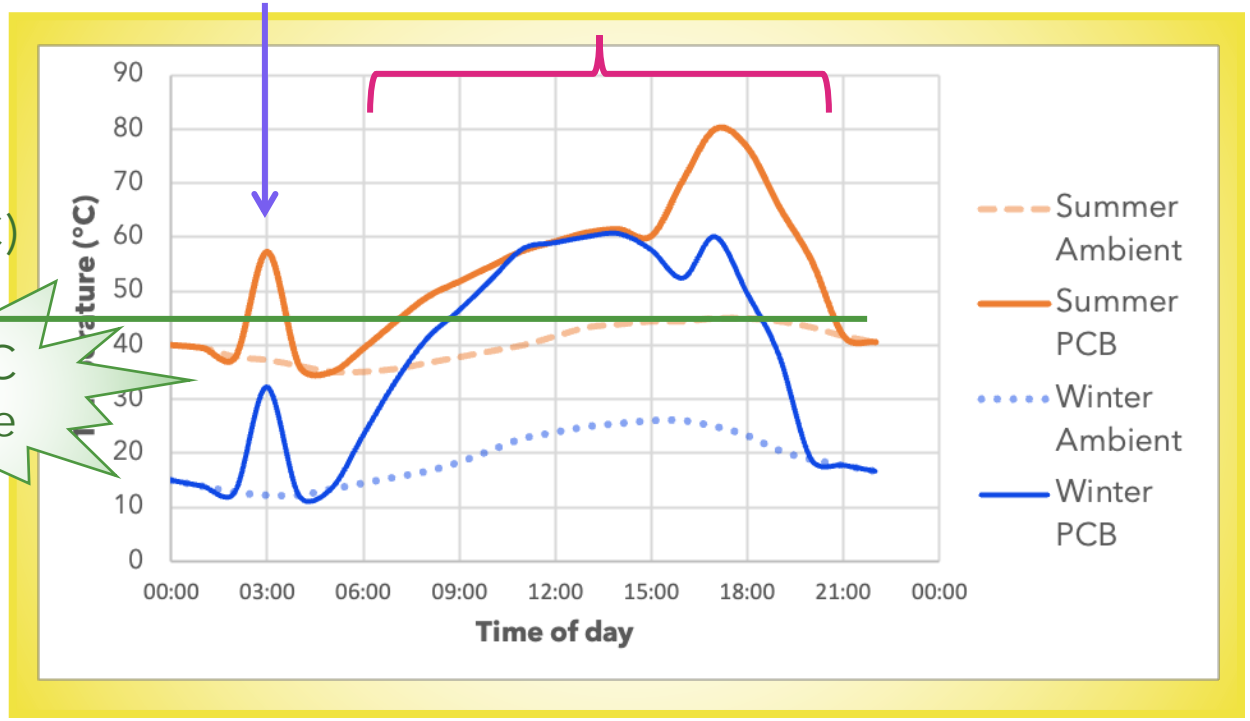
Define a Mission Profile

Ambient temperature



T_{app} (°C)

45°C average



Mission Profile

Inspired by: https://www.wunderground.com/history/daily/us/az/phoenix/KPHX/date/2023-7-16

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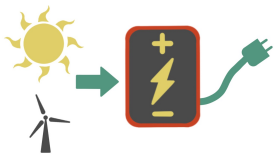
Correct our math



From Mission Profiles

	t_{app} (hours)	T_{app} (°C)	t_{bake} (days)
Example	12,000	87°C	58.2
Toy BMS	7,305	25°C	0.3
Car BMS	29,220	40°C	4.8
Solar backup BMS	175,320	**45°C**	43.4

Ok!

…with components rated to 125°C so accelerated lifetime tests can run in ~40 days.

t_{app} = 175,320 hrs

E_a = 0.7 eV

$$t_{bake} = t_{app} \times e^{\left(\frac{-E_a}{k_B} \cdot \left(\frac{1}{T_{bake}} - \frac{1}{T_{app}}\right)\right)}$$

k_B = 8.62x10^{-5} eV/K

T_{app} = 45°C

T_{bake} = 125°C

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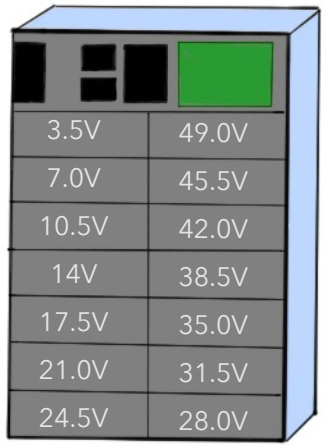
One last caveat for BMS mission profile



Some components are **always** connected to the cells. **Always** powered.

What does it mean to "be operating"?

- Exposed to voltage?
- Actively switching?

$24 \times 365.25 \times 8 = 70{,}128$ hrs

Car BMS Mission Profile

8 year lifetime

Used ~~10~~ hrs/day ← 2 hrs driving, 8 hrs charging

→ (10 × 365.25 × 8) = 29,220 hrs of operation over lifetime

Ave temp: T_{app} = 40 °C

$24 \times 365.25 \times 30 = 262{,}980$ hrs

Solar backup BMS Profile

30 year lifetime

Used ~~16~~ hrs/day ← 4 hrs discharging, 12 hrs charging

→ (16 × 365.25 × 30) = 175,320 hrs of operation over lifetime

Ave temp: T_{app} = 40 °C

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Key ideas

- Mission profile differs across applications
- Use **duration** and average **temperature** are inputs into Arrhenius equation
- Higher bake temperature **reduces** test time
- Higher mission profile temperature **increases** test time
- Higher bake temperature may require higher-temperature components

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Reliability

Video 1
- Introduction: Lifetime and accelerated lifetime testing
- Arrhenius equation
  - Review Arrhenius equation
  - Derive acceleration factor
  - AEC-Q100

Video 2
- Caveats of the approach
- Mission profile
- The design and test process

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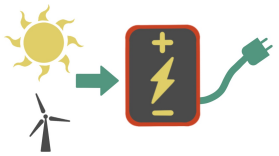
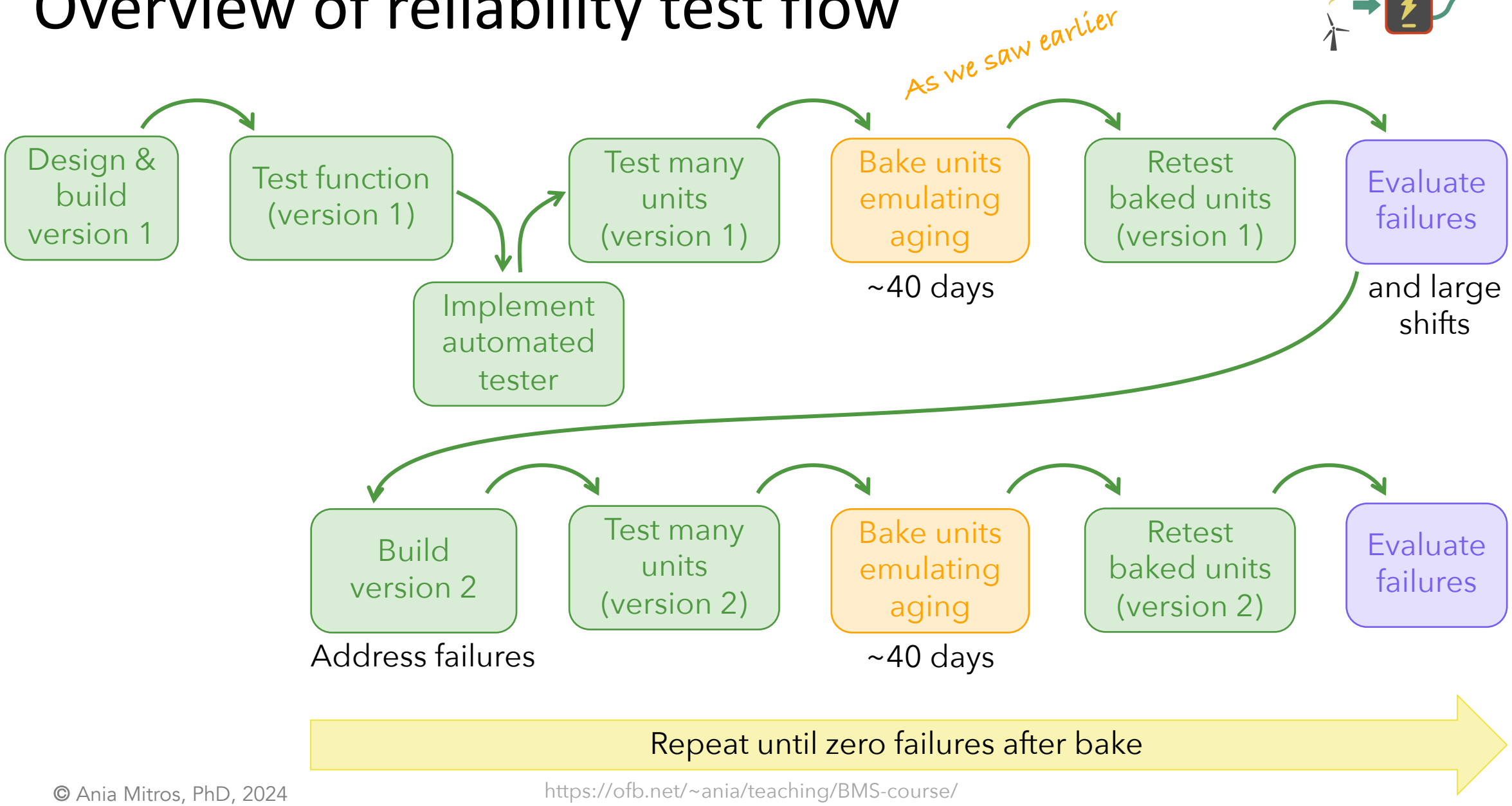
Overview of reliability test flow





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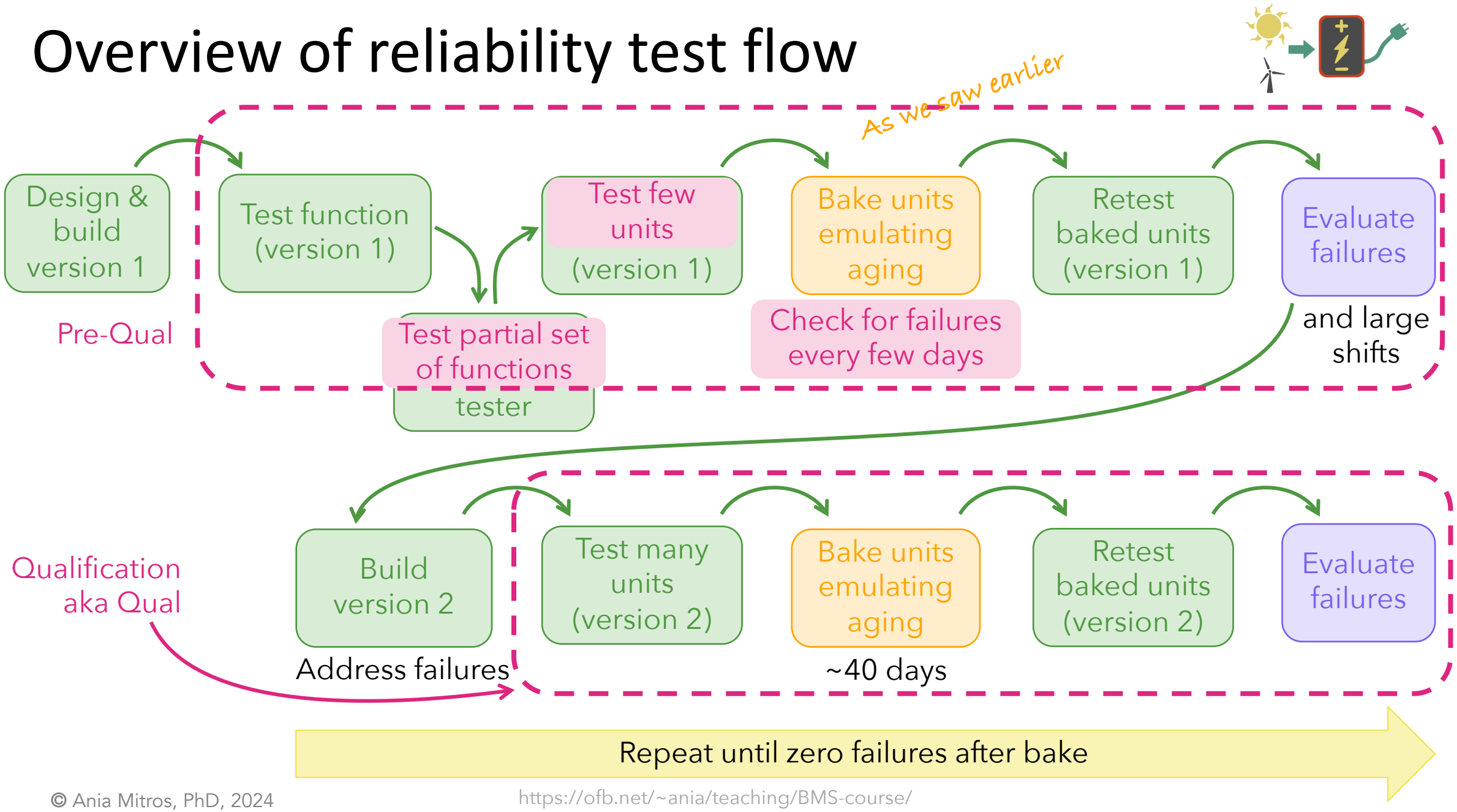
Overview of reliability test flow



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Design for Test & Reliability

Design & build version 1

Design for reliability from the start

- Use AEC-Q100 and AEC-Q200 components in non-automotive long-life applications
- Choose higher-temperature rated components (ex. 125°C parts for lower-temperature but long-life applications)
- Run pre-qual early, as soon as possible
- Consider self-heating and thermal design early
  - Ex. Large ground planes for hottest components
  - Ex. Thicker copper for ground plane

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The End

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